

System Architecture Documentation

This document describes the architecture, data flow, and key components of the real-time dashboard system.

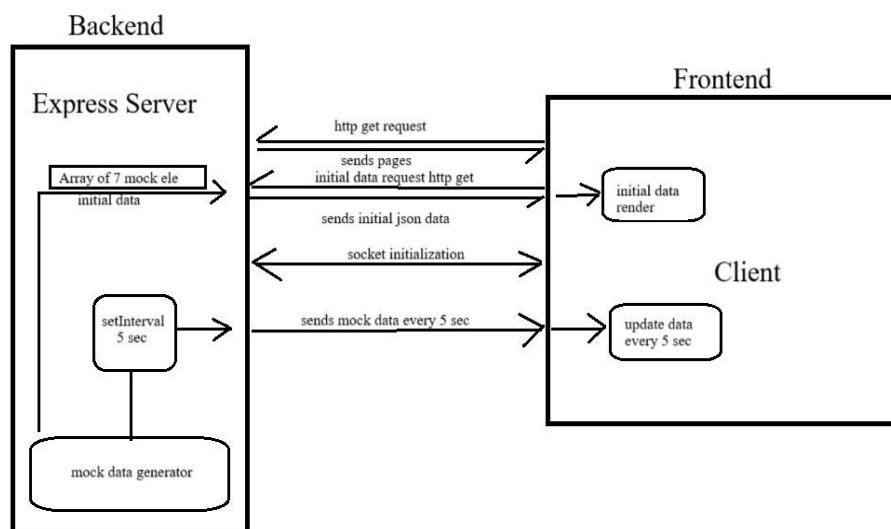
The dashboard uses a monorepo structure with React for the frontend, Express and Socket.IO for the backend, and mock data generation for real-time updates.

1. System Architecture Diagram

The system consists of the following components:

- **Frontend**: React (Client-side UI), where real-time data is displayed.
- **Backend**: Express, which serves the API and handles WebSocket connections using Socket.IO
- **Socket.IO**: Used for real-time bi-directional communication between the client and server.
- **Mock Data Generator**: Backend service that periodically generates random data for dashboard updates.
- **Database (optional)**: Not implemented in this version, but could store historical data.

Please refer to the System Architecture Diagram below (to be added manually):



2. Data Flow Explanation

Data flow is as follows:

- The **backend** generates mock data every 5 seconds and sends it via a WebSocket to the connected clients.
- **Frontend** (React app) listens for these WebSocket events and updates the UI by appending the new data to the dashboard in real-time.
- **Data Processing**: In the frontend, the data is processed and displayed with the current timestamp, allowing users to track live updates.

3. Tech Stack Breakdown

- **Frontend**:
 - React
 - Axios (for HTTP requests)
 - Socket.IO Client (for real-time communication)
 - Rechart
- **Backend**:
 - Node.js
 - Express (for creating the server)
 - Socket.IO (for real-time communication)
 - CORS (for handling cross-origin requests)
- **Data Flow**: Polling (Server generates data every 5 seconds) and WebSockets (for data transmission)

4. Real-Time Logic Flow

- **Polling**: The backend generates mock data every 5 seconds using `setInterval`.
- **WebSocket Communication**: Once the client connects to the server, the backend sends the data to all connected clients via WebSocket.
- **Frontend UI**: The frontend listens for new data and dynamically updates the dashboard without requiring page reloads.
- **Update Cycle**: Each update triggers a re-render of the React components displaying the new data.

5. Mock Data Design

- **Structure**: The mock data is a set of key-value pairs to simulate dashboard updates:

```
{ "timestamp": <current date and time in iso format> ,  
  "active_users": <Random integer between 20 – 99> ,  
  "page_views": < active_users * random no between 1-3> ,  
  "avg_session_duration": <Random Float between 1.5 -6.5> }.
```

- **Update Mechanism**: Every 5 seconds, new data is generated in the backend and broadcasted to clients through the WebSocket connection.

- **Data Format**: The data is structured as a JSON object with properties such as `value`, `timestamp`, and `status`. Each entry represents one piece of data that will appear on the dashboard.